line to that point of an adjacent curve which belongs to the same λ. Then (20) can be replaced by

$$(20a) \quad \begin{cases} \int_{\lambda_1}^{\lambda_2} \delta w \, d\lambda = 0 \\ w^2 = g_{\mu\nu} \frac{dx_\mu}{d\lambda} \frac{dx_\nu}{d\lambda} \end{cases}$$

However, since

$$\delta w = \frac{1}{w} \left\{ \frac{1}{2} \frac{\partial g_{\mu\nu}}{\partial x_\sigma} \frac{dx_\mu}{d\lambda} \frac{dx_\nu}{d\lambda} \delta x_\sigma + g_{\mu\nu} \frac{dx_\mu}{d\lambda} \delta \left(\frac{dx_\nu}{d\lambda} \right) \right\},$$

by substituting δw into (20a) and taking into account that

$$\delta \left(\frac{dx_\nu}{d\lambda} \right) = \frac{d \, \delta x_\nu}{d\lambda},$$

we obtain, after integration by parts,

$$(20b) \quad \begin{cases} \int_{\lambda_1}^{\lambda_2} d\lambda \, x_\sigma \, \delta x_\sigma = 0 \\ x_\sigma = \frac{d}{d\lambda} \left\{ g_{\mu\nu} \frac{dx_\mu}{d\lambda} \right\} - \frac{1}{2w} \frac{\partial g_{\mu\nu}}{\partial x_\sigma} \frac{dx_\mu}{d\lambda} \frac{dx_\nu}{d\lambda} \end{cases}$$

From this follows, due to the arbitrary choice of δx_σ, the vanishing of x_σ. Thus,

$$(20c) \quad x_\sigma = 0$$

are the equations of the geodesic line. If $ds = 0$ is not the case on the considered geodesic line, we can choose as parameter λ the “arc length” s measured along the geodesic line. Then $w = 1$, and one obtains in place of (20c)

$$g_{\mu\nu} \frac{d^2 x_\mu}{ds^2} + \frac{\partial g_{\mu\nu}}{\partial x_\sigma} \frac{dx_\sigma}{d\lambda} \frac{dx_\mu}{d\lambda} - \frac{1}{2} \frac{\partial g_{\mu\nu}}{\partial x_\sigma} \frac{dx_\mu}{d\lambda} \frac{dx_\nu}{d\lambda} = 0,$$

or by a mere change of notation

$$(20d) \quad g_{\alpha\sigma} \frac{d^2 x_\alpha}{ds^2} + \begin{bmatrix} \mu\nu \\ \sigma \end{bmatrix} \frac{dx_\mu}{ds} \frac{dx_\nu}{ds} = 0,$$

where, according to Christoffel, it is set that

$$(21) \quad \begin{bmatrix} \mu\nu \\ \sigma \end{bmatrix} = \frac{1}{2} \left(\frac{\partial g_{\mu\sigma}}{\partial x_\nu} + \frac{\partial g_{\nu\sigma}}{\partial x_\mu} - \frac{\partial g_{\mu\nu}}{\partial x_\sigma} \right).$$

Finally, if one multiplies (20d) by $g^{\sigma\tau}$ (outer multiplication with respect to τ, inner with respect to σ), one obtains at last the final form of the equation of the geodesic line